Holograms

A hologram is a 3-D image made using laser beams. It is stored inside a photographic film as an interference pattern, which contains information about the object's surface. The image you see when you view a hologram has depth, and by moving your head, you are able to look at it from different angles.

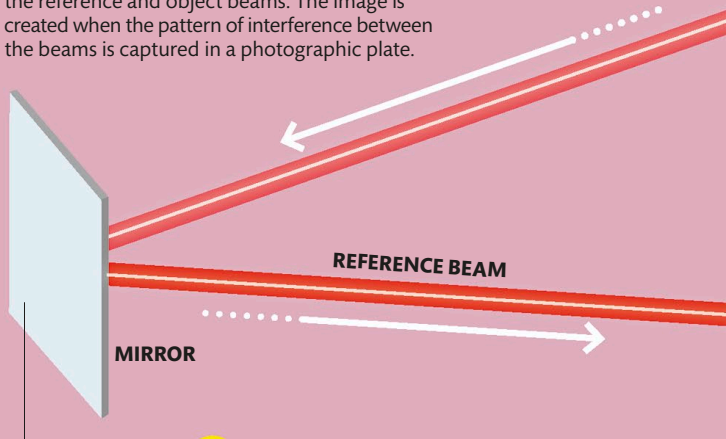
ARE THE HOLOGRAMS OF MUSICAL PERFORMERS AT CONCERTS REAL HOLOGRAMS?

No, they are images created by mirrors—an illusion called "Pepper's Ghost."

Making a hologram

Holograms are made using laser light. Significantly, the light waves produced by a laser are all "in step" (see pp.146–147). To make the hologram, a laser beam passes through a splitter. Half of the light forms the reference beam, which is directed straight onto a photographic (light-sensitive) film. The other half forms the object beam, which bounces off the object to be pictured in the hologram. The reflected object beam falls onto the film, where its waves merge, or interfere, with the reference beam. The interference creates a pattern that holds information about the surface of the object—and this information can be extracted if light shines onto the film once it has been developed.

Holographic image
A hologram is formed when two beams combine: the reference and object beams. The image is created when the pattern of interference between the beams is captured in a photographic plate.



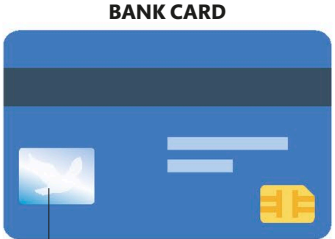
Mirror reflects reference beam onto diverging lens

3 Reference beam
Light that passes through the beam splitter bypasses the object. It reflects off another mirror that directs it toward the photographic film. First it passes through a diverging lens that broadens out the beam.

IF YOU BREAK A HOLOGRAM INTO FRAGMENTS, EACH FRAGMENT CONTAINS THE WHOLE IMAGE

SECURITY HOLOGRAMS

The holograms on banknotes, credit cards, and concert tickets are designed to prevent those items from being forged. They are produced with laser beams but are visible in ordinary white light.



Viewing a hologram

The hologram described above is called a transmission hologram. Another kind is a reflection hologram. It is similar, but there is no beam splitter: the reference beam passes through the film then reflects off the object, which is positioned behind the film, to become the object beam. When the photographic film is developed, it looks dark, with strange lines on it; there is no sign of an image. To view a reflection hologram, a laser beam passes through the film, reflecting off the interference pattern inside it and producing an image.